

# Physoc Code Tutorials README

Welcome! This folder contains all the materials from the Physoc tutorial session earlier this semester. It's designed to help you get started with LaTeX and Python, two powerful tools for physics students, whether you attended the session or are working through it on your own. What's inside:

1. Slides: a pdf copy of the slides from that session
2. A .zip file called `Physoc_LaTeX_Tutorial.zip`
3. A folder called `Physoc_Python _Tutorial`
4. This document (a README): A quick explanation of what's included and how to use it.

## 1 LaTeX Tutorial

This tutorial is provided as a .zip file because that is the format Overleaf prefers for uploading entire projects. If you're using Overleaf, you do **not** need to extract the contents, simply upload the whole file as-is. If you're using a different LaTeX editor (such as TeXstudio or VS Code), you may need to extract the files and open the main .tex document manually.

The zip file contains an example LaTeX project that walks you through the basics of writing structured physics reports. It includes a sample document with comments, explanations, and blank sections for you to complete as practice.

The activities we worked through during the session included:

1. Replace the images a b and c with something old, something new, and something blue
2. Add two references written by a Glasgow university researcher in the past 2 years. How can you search?
3. Write out and reference the higgs boson potential formula:

$$U(\phi) = \frac{1}{4}u_0 \left( |\phi|^2 - \lambda^2 \right) \quad (1)$$

This tutorial is designed to help you build confidence using LaTeX for coursework, lab reports, and future research projects.

## 2 Python Tutorial

This folder contains a Jupyter notebook and a .csv file. You will need both to complete the activities. The notebook is provided in two versions:

- An **empty version**, where you can write your own code and work through the tasks independently.
- A **completed version**, which includes example solutions and explanations.

We recommend starting with the empty version and trying to solve each activity yourself before checking the answers. This will help you build confidence and get used to writing Python code from scratch.

The tutorial covers basic data handling and plotting using `numpy` and `matplotlib`, with examples relevant to physics coursework. It's designed to be beginner-friendly and self-paced, so take your time and experiment with the code.